

Pi Zero Portable BILINGUAL TRANSLATOR

SOMNATH BERA

This system is designed as a bilingual translator, leveraging the gTTS library to support multiple Indian languages, including English (en), Bengali (bn), Gujarati (gu), Hindi (hi), Kannada (kn), Malayalam (ml), Marathi (mr), Tamil (ta), Telugu (te), and Urdu (ur). The two-letter codes in brackets specify the target language for speech conversion. For example: `gTTS(text = translated_text, lang = dest_lang)`. Speech conversion between languages is straightforward—simply adjust the source and target language codes. For instance, Gujarati speech can be translated into Spanish by updating the relevant codes.

Additionally, Google Text-To-Speech (gTTS) supports numerous international languages, such as Arabic (ar), Chinese Simplified (zh), Chinese Traditional (zh-tw), Czech (cs), Danish (da), Dutch (nl), English (en), Filipino (fil), Finnish (fi), French (fr), German (de), Greek

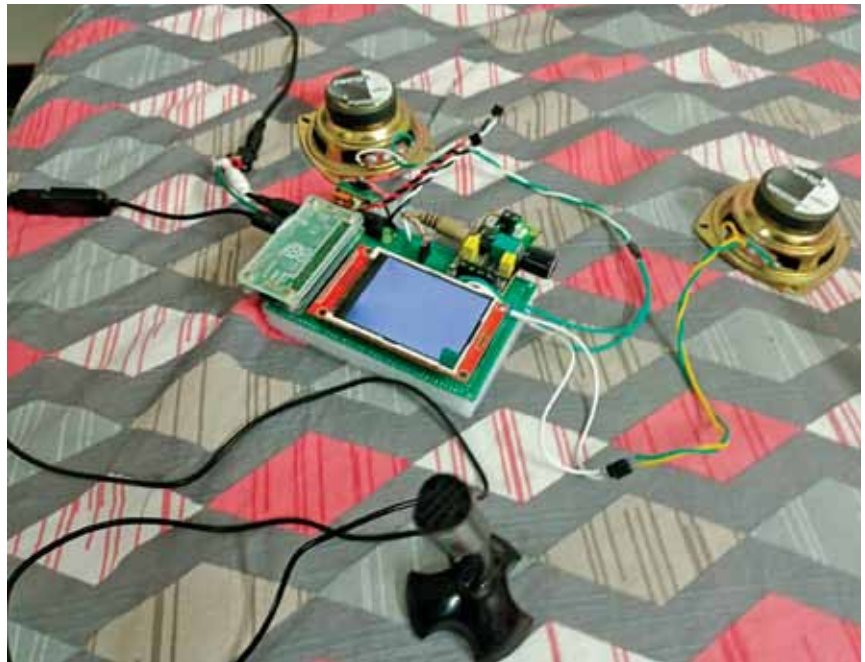


Fig. 1: Author's prototype

(el), Hebrew (he), Hindi (hi), Hungarian (hu), Indonesian (id), Italian (it), Japanese (ja), Korean (ko), Norwegian (no), Polish (pl), Portuguese Brazil (pt-br), Portuguese Portugal (pt), Russian (ru), Spanish (es), Swedish (sv), Thai (th), Turkish (tr), Ukrainian (uk), Vietnamese (vi), and others. The author's prototype, featuring a 240x320 display and 5W + 5W amplifiers, is shown in Fig. 1.

The testing covered English (en), Gujarati (gu), Bengali (bn), and Hindi (hi) for Indian languages, alongside Russian (ru), Spanish (es), and German (de) for international languages. So, apart from English, Bengali, Gujarati, and Hindi, the accuracy of the outputs in

other languages cannot be verified without additional feedback from expert readers.

Configuration

The device requires a Raspberry Pi Zero installed with the latest Raspbian OS. Python 3 must be installed, along with the required libraries. These can be installed via the Linux terminal using the following commands:

X. Install python & pip:

```
$> sudo apt install python3 python3-pip
```

1. Install gTTS (Google Text-to-Speech):

```
$> pip install gTTS
```

2. Install SpeechRecognition:

```
$> pip install SpeechRecognition
```

3. Install Adafruit-GPIO and

BILL OF MATERIALS

Components	Quantity
Raspberry Pi Zero board (MOD 1)	1
SPI TFT display (MOD 2)	1
PAM8403 module (MOD 3)	1
Switch (S1, S2)	2
Resistors 270Ω (R1, R2)	2
Resistors 150Ω (R3, R4)	2
10μF electrolytic capacitors, 16V (C1, C2)	2
0.1μF ceramic capacitors (C3, C4)	2
5V DC, 2A power supply adaptor	1
Speaker 5W	2
USB microphone1	1